TEXORA · LECTURE SHEET

Protective and Safety Items

The special-apparel catalogue organised by hazard — heat and flame, molten splash, chemicals, mechanical injury, visibility, static, cold, rain and biological risk — with the garment constructions, fibres and performance standards that answer each hazard.

AM-4103 · CLOTHING COMFORT & SPECIAL APPAREL PRODUCTS · TOPIC 3 · LEVEL 4 TERM 1

1. Protection is a system specification

A protective garment is specified backwards from a **risk assessment**: what hazard, at what intensity, for how long, on which body parts — then matched to a standard whose test reproduces that hazard (Topic 4). Protective clothing is regulated PPE in most markets (e.g. EU Regulation 2016/425), certified by notified bodies and marked with the standard number and performance levels: the label is part of the product. The recurring design tension of this whole course is visible here in its sharpest form: **protection and comfort pull against each other** — every barrier added against a hazard also resists heat and sweat escape, so protective design is the art of meeting the standard at the lowest physiological cost (Topics 1–2 parameters are the cost accounting).

2. The catalogue by hazard

HAZARD FAMILY	TYPICAL ITEMS	MATERIALS / CONSTRUCTION	KEY STANDARDS
Heat & flame	Foundry and welding clothing, firefighter turnout gear, racing suits, industrial FR workwear	Aramids (meta-/para-), FR-treated cotton, modacrylic blends; multilayer turnout systems: outer shell + thermal liner + moisture barrier; no melt-drip fibres next to skin.	EN ISO 11612 (heat/flame), EN ISO 11611 (welding), EN 469 (firefighters)
Molten metal splash	Smelter jackets, trousers, spats, gloves	Wool and FR fabrics with smooth shed surfaces; closures covered; pockets external or absent so splash cannot lodge.	EN ISO 11612 codes D (aluminium) and E (iron) splash levels
Chemicals	Chemical suits (Types 3–6), laboratory and	Coated/laminated fabrics, welded or taped seams; permeation resistance chosen per	EN 14605 (Types 3/4), EN ISO

	agrochemical clothing	chemical (breakthrough time is the rating).	13982 (Type 5), EN 13034 (Type 6)
Mechanical	Cut-resistant gloves and sleeves, chainsaw trousers, stab/ballistic vests	HPPE, para-aramid and steel-core yarns; chainsaw clothing uses layered fibres that clog the sprocket; ballistic packs of woven aramid or UHMWPE.	EN 388 (gloves), EN ISO 11393 (chainsaw), NIJ standards (ballistic)
Visibility	Hi-vis vests, jackets, trousers for road/rail/construction	Fluorescent background fabric (day) + retroreflective tape (night) in prescribed minimum areas and design (bands encircling torso/limbs).	EN ISO 20471 (classes 1–3)
Electrostatic	ESD and ATEX-zone garments, cleanroom wear	Conductive fibres (carbon core, metal) in a grid; dissipative fabric; garments must earth the wearer, seams and closures included.	EN 1149 (protective), EN 61340 (ESD)
Cold & rain	Cold-store clothing, offshore and foul-weather sets	High-loft insulation with windproof shell (cold); waterproof-breathable laminates with taped seams (rain) — Ret class is the comfort rating.	EN 342 (cold), EN 343 (rain)
Biological / medical	Surgical gowns, isolation gowns, coveralls	SMS nonwovens and film laminates; barrier graded by liquid/viral penetration tests; single-use systems dominate.	EN 13795 (surgical), EN 14126 (infective agents)

3. High-visibility design: a worked specification

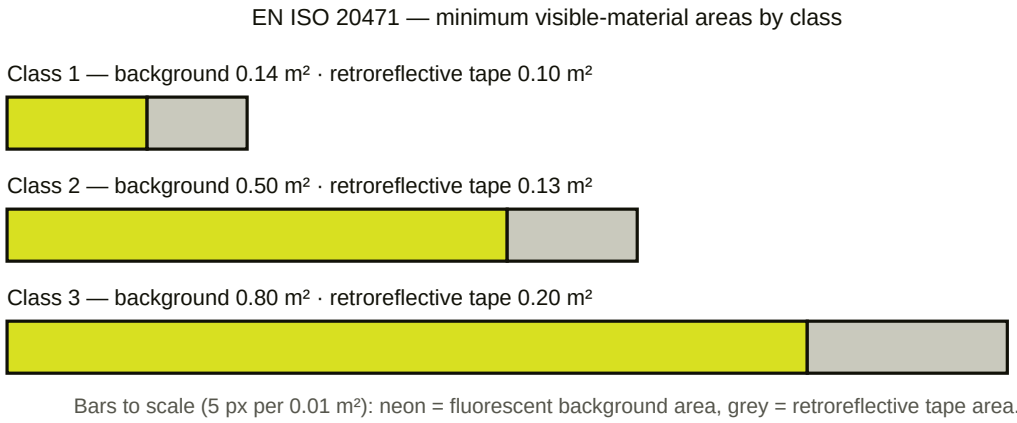


Figure 1 — Hi-vis classes are, in the first instance, an area specification: the garment must present at least these areas of fluorescent and retroreflective material. Class is set by area AND garment design rules, not area alone.

The class logic generalises across the catalogue: a **performance level printed on the label** (letter codes in EN ISO 11612 — e.g. limited flame spread A, convective heat B, radiant heat C, molten splash D/E, contact heat F — each with graded levels) tells the buyer exactly which tests the garment passed and how well. Specifying protective clothing means writing those codes into the order, not writing "fireproof jacket". Note also what the codes imply for comfort: an EN ISO 11612 D3 aluminised foundry system will have a very high Ret — the wearer's heat stress, work/rest cycles and hydration become part of the safety system, decided with Topic 5's heat-balance arithmetic.

4. Fibres behind the protection

FIBRE / MATERIAL	PROTECTIVE CONTRIBUTION	LIMITS TO RESPECT
Meta-aramid	Heat and flame resistance, no melting; char forms a barrier.	Degrades under UV; expensive; modest strength.
Para-aramid	Very high strength and cut resistance; ballistic and chainsaw layers.	UV-sensitive; poor compression strength.
FR-treated cotton	Durable flame retardancy at workwear price; comfortable hand.	Finish durability depends on laundering discipline; heavier when wet.
Modacrylic / FR viscose blends	Inherent FR with soft handle; blends tune cost and comfort.	Lower strength; blend ratios matter to performance.
HPPE / UHMWPE	Cut resistance and ballistic energy spread at low weight.	Melts at ~150 °C — never the answer to a heat hazard.
PTFE / PU membranes	Waterproof-breathable barriers; chemical splash barriers in laminates.	Ret penalty; seam sealing is the weak point, not the film.

Glossary

TERM	MEANING
PPE	Personal protective equipment; regulated product category requiring certification and marking.
Performance level / code	The graded result printed for each test in a product standard (e.g. 11612 code C2).
Breakthrough time	Time for a chemical to permeate a barrier fabric at the standard's detection rate; the rating for chemical clothing.
Retroreflective	Returning light to its source (vehicle headlights → driver), the night-time half of hi-vis.
Turnout gear	Firefighter ensemble: outer shell, thermal barrier and moisture barrier worn as a system.

Key points and summary

- Protective items are specified from a risk assessment to a named standard and printed performance levels — the label is part of the product.
- Each hazard family has its fibre logic: aramids for heat/cut, laminates for chemicals, area rules for visibility, conductive grids for static.
- Protection is bought with comfort: barrier performance raises Ret and heat stress, which must be managed as part of safety.
- EN ISO 20471 classes quantify hi-vis by minimum areas of fluorescent (0.14/0.50/0.80 m²) and retroreflective (0.10/0.13/0.20 m²) material.

Practice

Answer in your own words; brief model answers follow.

1. A road crew works at night beside fast traffic. Which hi-vis class do you specify and why not a lower one?
2. Why is HPPE an excellent cut fibre but a dangerous choice for a foundry glove?
3. A chemical suit passes its fabric permeation test but leaks in wear. Where is the failure most likely, and which construction details control it?
4. Explain, with the course's parameters, why a fully certified FR garment can still be a safety problem in a hot plant.

5. What two different visibility mechanisms does a hi-vis garment combine, and when does each work?

Answer key

1. Class 3: night work beside fast traffic is the highest-risk visibility case; Class 3 requires the largest fluorescent area (0.80 m²) and tape area (0.20 m²) with torso-encircling design, maximising both daytime conspicuity and night retroreflection. Lower classes are specified for lower-risk, slower environments.
2. HPPE's high tenacity and low weight give top cut scores, but it melts at about 150 °C: in a foundry the glove itself becomes a molten-polymer burn hazard. Heat hazards need fibres that char or resist heat (aramid, FR leather/wool systems), not thermoplastics.
3. At the seams and closures: stitch holes, untaped seams, zips and interfaces (glove/collar) bypass the fabric barrier. Welded/taped seams, covered closures and system testing of the finished garment control it — fabric data alone never certifies a suit.
4. FR performance says nothing about heat strain. The same barrier stack raises Ret (Topic 2), so metabolic heat and sweat cannot escape; core temperature climbs (Topic 5), and heat illness — itself a serious safety event — arrives sooner. Work/rest cycles, hydration and the lowest-Ret construction that still meets the FR codes are part of the specification.
5. Fluorescence (daylight: converts UV to visible light, conspicuous against backgrounds) and retroreflection (night: returns headlight beams to the driver). One needs daylight, the other a light source near the observer's eye line.

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